

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Let's talk about Software Engineering



Learning Outcome:

Identify information about the history of software and its characteristics in simple spoken and written situations.
Give information about software history, characteristics and methodologies.

Can Do Statements:

- I can give information about the origins of software engineering.
- I can talk about the environment that originated software companies.
- I can present information about the transformation of software companies.

Ask your partner:

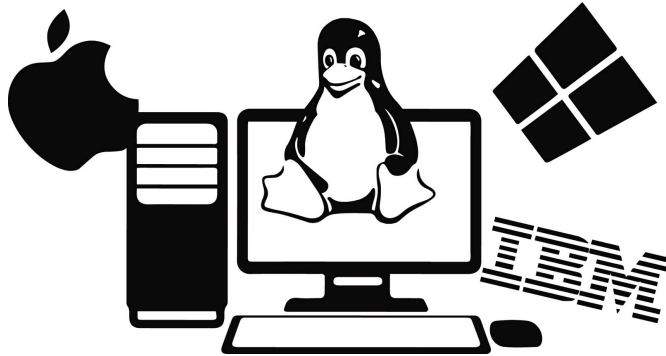
- Why do you think software development became important?
- What software development companies do you know? Which one do you think is the most important?





Let's Explore!

Maurice Smith is explaining the origins of software engineering!

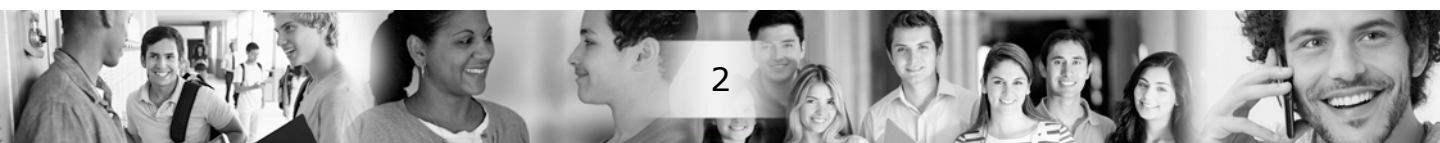


Instructions: Read the text below.

Modern computer history starts in April 1964 when IBM releases the System/360. It was the first family of computers that worked under the same architecture. This was a revolutionary invention as before, each computer had its own peripherals and applications. For achieving this new family of computers, there was a very important task that software developers had to carry out: They had to create an operating system that worked for all S/360s. The result was the OS/360. This was a very innovative software that Fred Brooks, the father of software engineering, developed with a big group of programmers.

There is another important event in the history of software engineering: the Garmisch conference in Germany. Thanks to this conference, people started to consider software development as a science and not as an art. After this, there was a change in the field of software development and it was the first time that people used the term 'engineering' to refer to this field. Now, it was a more structured discipline and new methodologies for software development appeared. In addition, another important event took place in 1968 when IBM decides to unbundle its software. Before, there were computers that came with their own specific software, but since IBM changed the rules, companies sold pieces of software that were compatible with different computers.

This originated the environment for new companies and independent software vendors to start selling their own pieces of software. Before, there were few companies that developed software, but after IBM unbundled its software and as computers became more popular, many new companies started developing software packages that were compatible with IBM computers. This marks the beginning of the mass production of software. There were different companies that specialized in the development of particular applications or operating functions. As there were many small companies, most of them ended merging with each other to form bigger companies or groups of vendors.





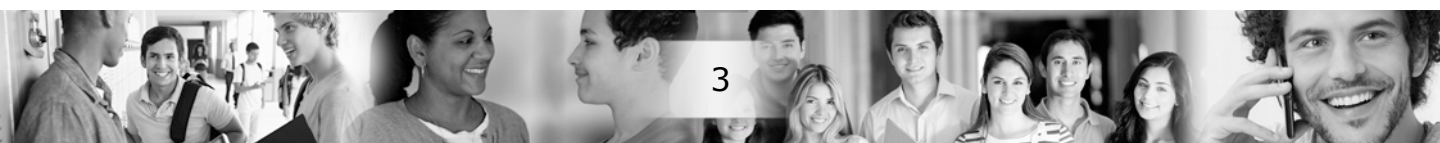
Let's Practice!

The Origins of Software Engineering

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

1. There was a seminal conference that took place in Garmisch, Germany in 1968 called "Software Engineering".
2. There were few companies developing software before IBM unbundled its software.
3. There was the idea that software was an art and not a science.
4. There was a popularization of the term computer engineering in the 1960s.
5. There were many companies selling their pieces of software in the mid-1970s.
6. There were important changes in perceptions about hardware and software fields after the Garmisch conference.

Sentences With the Singular Form of 'There Be'	Sentences With the Plural Form of 'There Be'



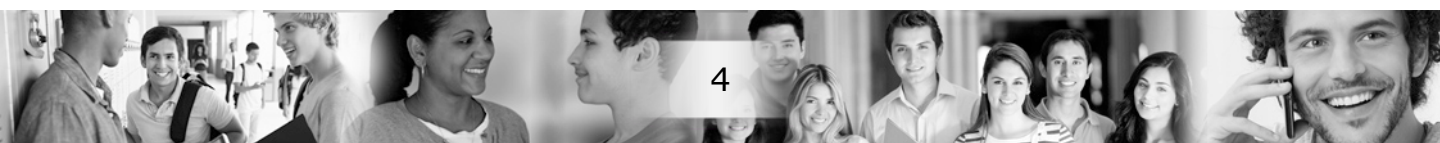


Let's Practice!

When the Software Market Began to Increase

Instructions: Select the correct answer.

1. These two events marked the beginning of the mass production of software:
 - a. IBM manufactures military equipment and develops new applications.
 - b. IBM unbundles its software and computers become more popular.
 - c. New architecture developments appear and computers become more popular.
 - d. IBM unbundles its software and manufactures military equipment.
2. Besides the Garmisch conference, what is the other event that helped the development of the software industry?
 - a. The Cold War
 - b. IBM was created as a software company.
 - c. When IBM unbundled its software.
 - d. Independent vendors take control of the market.
3. In 1968, IMB started to sell...
 - a. franchises more often.
 - b. mainframes.
 - c. hardware to China.
 - d. its pieces of software separately.
4. Once IBM unbundled its software...
 - a. the S/360 appears in the market.
 - b. people stop buying IBM computers and they start supporting new emerging companies.
 - c. new companies and independent software vendors start selling their own pieces of software.
 - d. the government sues the company.
5. What is the name of the operating system that worked for all S/360?
 - a. OS/360
 - b. O/360
 - c. OS/390
 - d. S/390
6. What happens in the Garnisch conference that helped change the perception of software development?
 - a. People became more interested in the field of software development.
 - b. People started to see software development as a science and not as an art.
 - c. People started talking more about the mass production of software.
 - d. People see software development not as a science but as a form of art.





Extension Activity

History of Software Development: Timeline

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about the history of software development.
- Design a timeline with three to six important events that marked the history of software development.
- Show the timeline to your partners.

Self-Assessment

Yes

No

- I look for information about the history of software development.
- I design a timeline of the history of software development.
- I explain the timeline to my classmates.

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Target Words

- **compatible** (adjective): that works with different devices, that is read by different devices
e.g. Before, there were computers that came with their own specific software, but since IBM changed the rules, companies sold pieces of software that were compatible with different computers.
- **unbundle** (verb): to separate charges for products or services usually offered as a package
e.g. Another important event takes place in 1968 when IBM decides to unbundle its software.
- **mass production** (noun): the manufacturing of big amounts of a specific product
e.g. After IBM unbundled its software and as computers became more popular, many new companies started developing software packages that were compatible with IBM computers. This marks the beginning of the mass production of software.
- **take place** (idiom): to occur or happen
e.g. Another important event took place in 1968 when IBM decides to unbundle its software. Before, there were computers that came with their own specific software, but since IBM changed the rules, companies sold pieces of software that were compatible with different computers.
- **innovative** (adjective): introducing something new
e.g. This was a very innovative software that Fred Brooks, the father of software engineering, developed with a big group of programmers.
- **merge** (verb): to get mixed with a different group of elements, to mix two or more elements into the same category
e.g. As there were many small companies, most of them ended merging with each other to form bigger companies or groups of vendors.



Learning Points

There Be in Simple Past Tense Affirmative Statements

You use **There was** or **There were** in **affirmative** to say that something existed in the past.

We use **There was** or **There were** when we refer to the existence or presence of someone or something in the past.

You can use **There was** for singular and uncountable nouns, and **There were** for plural countable nouns.

Affirmative form	Example:
Singular noun (woman)	<i>There was a woman at the door.</i>
Plural noun (women)	<i>There were women at the door.</i>
Uncountable noun (software)	<i>There was money on the desk.</i>

Examples

- *Before, there were computers that came with their own specific software, but since IBM changed the rules, companies sold pieces of software that were compatible with different computers.*
- *As there were many small companies, most of them ended merging with each other to form bigger companies or groups of vendors.*
- *For achieving this new family of computers, there was a very important task that software developers had to carry out: They had to create an operating system that worked for all S/360s.*



Learning Points

Adjectives

Adjectives are words that describe or modify nouns. There are 2 basic positions for adjectives: before a noun and after some verbs. For this learning point, we are going to learn about adjectives that go after the verb **to be**.

For most cases, when we want to describe a noun, we place the adjective before that noun. However, when we use the verb **to be**, we place the **adjective** after the verb. Take a look at the next example. Here, the verb **to be** is followed by an **adjective** that describes the state of the group of people **"they"**.

Example:

*They were **happy** when they worked in the software project.*

Now, when we want to place a noun after the adjective, we have to start the sentence with a pronoun (there, this, it, etc.) that refers to the noun we are going to mention.

Examples:

*There is an **important** concept we need to consider.*

*It is a **wonderful** idea!*

*Those were very **useful** pieces of software.*

In these examples the pronouns *there*, *it* and *those* refer to the nouns that come after the adjectives *concept*, *idea* and *pieces of software*, respectively.

Examples

- *Modern computer history starts in April 1964 when IBM releases the System/360. It was the first family of computers that worked under the same architecture.*
- *This was a revolutionary invention as before, each computer had its own peripherals and applications.*
- *Now, it was a more structured discipline and new methodologies for software development appeared.*